"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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(3) a) $H^2 = H.H = (I-2uu^T)(I-2uu^T) = I^2 - 1.2uu^T - 2uu^T.1 + 4uu^T.uu^T$

$= I(I-2uu^T-2uu^T) + 4u.\|u\|^2.u^T$

$= I - 2uu^T - 2uu^T + 4u.\|u\|^2.u^T$ yani $\|u\|^2 = u^T.u$

i.e $\|u\| = \pm\sqrt{u^T u}$

$= \|u\| = 1$

yani $H^2 = I - 4uu^T + 4uu^T$

$H^2 = I - 4(1) + 4(1)$

$H^2 = I$

b) $(I-2uu^T).(I-2uu^T)^T = (I-2uu^T).(I^T-(2uu^T)^T)$

$= (I-2uu^T).(I-2(u^T)^T u^T) = (I-2uu^T).(I-2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T uu^T = I - 4uu^T + 4uu^T$

$= I - 4(1) + 4(1) = H.H^T = I$

$(\because (A-B)^T = A^T - B^T \quad (\because (AB)^T = B^T.A^T \quad (\because \|u\| = 1$

$H^T H = I$

yani, $HH^T = H^T H = I$ $\qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

bundan dolayı H onthogonal matrix

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3) d) $H=(I-2uu^T)$ verilmiştir $H^T=(I-2uu^T)^T$ $(\because (A-B)^T=A^T-B^T$

$=I^T-2(uu^T)^T$ $=I-2(u^T)^T u^T=I-2uu^T=H$ $\because ((AB)^T=B^T.A^T)$

yani H simetrik bir matristir. Ve şunları gösteren bir teorem var:

$H=\Phi\Lambda\Phi^T$ (Φ orthogonaldir)

$=\Phi\Lambda\Phi^{-1}$ ($\Phi\Phi^T=I=\Phi^T=\Phi^{-1}$)

$=(\Phi^{-1})^{-1}\Lambda\Phi^{-1}$ (Λ = diagonal matristir)

$=P^{-1}\Lambda P$

yani H diagonalizable matristir.

1) a) $x_2+x_4=0$ $-\frac{1}{2}x_3+x_4=0$ $x_2=x_3$

$k=-\frac{1}{2}$

$mx_1+x_3=0$ $-8x_1+x_2=0$

$mx_1=-x_3$ $x_2=8x_1$

yani, $x_2=x_3$ ve x_2 $8x_1$ ve $-mx_1=x_3$

$-mx_1=8x_1$ $m=-8$

c) $\begin{bmatrix}1&0&0\\0&1&0\\0&-1&1\\0&-\frac{1}{2}&0\end{bmatrix}=1$

$\begin{bmatrix}1&0&0&0\\-8&1&0&0\\-8&0&1&0\\-1&0&0&1\end{bmatrix}=1$

$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&-\frac{1}{2}&1\end{bmatrix}=1$

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3) c) For a matrix A, a vector (non zero) will be eigen vector (supposed) of A corresponding eigen value if,

$Av = \lambda v$

$= (I - 2uu^T).u$

$= Iu = 2uu^Tu$

$= u - 2u.\|u\|^2$ as F is a identity matrix, so I is the only eigen value I and u is unit vector so $\|u\| = 1$:

$+\sqrt{u^Tu}$ u is $= \lambda = -1$

4)

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